Knowledge, Practices, and Challenges of Infection Control Among Nurses in Tertiary Healthcare Settings

Introduction

Hospital acquired infections remain a significant challenge in healthcare worldwide. Patients sometimes contract new infections during treatment, leading to worse outcomes and extended recovery times. These infections affect both patients and healthcare workers, particularly nurses who maintain close patient contact.[1] In tertiary care hospitals handling complex cases, immunocompromised patients, invasive devices, and high patient turnover create environments where infectious agents spread more easily.[2]

Standard infection control measures address these risks. Hand hygiene remains the cornerstone of prevention, along with proper use of gloves and masks, safe needle disposal, regular cleaning, and equipment sterilization.[3] International health organizations have published detailed guidelines. Yet compliance often falls short. Studies document a persistent gap between theoretical knowledge and actual practice.[1,4] Nurses might correctly answer questions about protocols while admitting they skip proper technique during busy shifts.

Pakistan's healthcare system faces pressures that compound compliance challenges. Public hospitals operate with nurse to patient ratios below recommended levels. Supply issues leave wards without adequate protective equipment. Administrative oversight remains inconsistent.[2,5] A nurse working a 12 hour shift with twice the recommended patient load, facing glove shortages, and receiving no feedback exists in a different context than controlled settings where protocols are developed. Tertiary hospitals treating the sickest patients represent settings where prevention failures carry the highest stakes, yet research examining these facilities remains limited.

Gaps in Current Research

Existing studies from Pakistan document the knowledge practice disconnect, but important questions remain unanswered. Most research measures knowledge and practice separately without examining what prevents nurses from acting on their knowledge. Survey methods dominate, creating potential for response bias.[1-5] Few studies employ observational methods or examine actual compliance through direct monitoring.

The specific context of tertiary hospitals has received insufficient attention. These facilities handle referral cases, treat complex conditions, and maintain intensive care units where infection risks are elevated. Whether barriers differ between tertiary and district hospitals remains unclear. Research has not explored variation within hospitals either. Most importantly, studies treat structural constraints as background factors

rather than central inquiry subjects. When international guidelines assume consistent access to running water and unlimited sterile gloves, how do nurses in resource limited settings prioritize competing demands?

Study Objectives

This research examines infection control among nurses in tertiary care settings through four objectives:

- Assess current knowledge levels regarding infection prevention protocols
- Evaluate reported infection control practices during patient care
- Identify institutional, environmental, and individual barriers impeding adherence
- Analyze relationships between knowledge, practices, and barriers

Review of Related Literature

Khowaja and colleagues examined a teaching hospital in Thatta, finding nurses demonstrated satisfactory understanding of principles but actual compliance fell below acceptable standards. The study identified inadequate continuing education and irregular protective equipment availability as key obstacles.[1] However, it did not explore how these barriers interact. Would guaranteed glove supply improve compliance, or do multiple factors need simultaneous change?

Batool documented concerning practices in Quetta including insufficient hand hygiene and unsafe needle handling. Environmental factors emerged as significant contributors, particularly poor sanitation and absent monitoring.[2] Individual behavior cannot be separated from institutional culture. When hospital wide standards remain lax and no one monitors compliance, maintaining personal adherence becomes difficult.

Kousar's team studied public hospitals, reporting high awareness but inconsistent implementation. Their analysis emphasized infection control committees and surveillance programs.[3] While these align with international practices, the study did not examine whether such systems actually function in Pakistani hospitals. Many facilities may have official committees that rarely meet.

Soomro examined Isra University Hospital, particularly intensive care units where risks are most severe. They found poor hand hygiene, inadequate supplies, and excessive workload.[4] High acuity areas present distinct challenges. Nurses perform invasive procedures and work with vulnerable patients. Whether these areas need specialized interventions beyond general programs remains unclear.

Baqar documented occupational hazards from blood exposure in Khairpur. Poor compliance increases risks for workers themselves.[5] Yet the study did not investigate

why nurses accept these risks. Do they underestimate dangers? Do time pressures force shortcuts? Understanding nurse perceptions could inform better training.

Synthesis of Evidence

Common themes emerge across studies. Knowledge levels appear generally adequate. Most nurses can identify proper procedures when asked. Yet practice consistently falls short. This pattern holds across different cities and hospital types. The explanations involve resource constraints, workload, and training gaps. However, current research describes barriers without deeply analyzing them. How much does workload need to decrease before compliance improves? Few studies provide this detail. Moreover, most rely on self reported data, which likely inflates compliance estimates.

Tertiary facilities receive less attention than warranted. These hospitals treat referral cases, perform complex procedures, and train future workers. Yet few studies examine them specifically. Finally, the literature has not adequately addressed resource scarcity realities. International guidelines assume certain baseline resources exist. When these assumptions don't hold, how should protocols be adapted? This study addresses these gaps by examining knowledge, practices, and barriers together in the tertiary hospital context.

References

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